

AGT-44 Internal Maintenance Limits Reference

Internal maintenance-limit register for AGT-44 reactor agitator gearbox and mechanical seal records. The tables define same-unit comparison limits and record-handling rules for site maintenance documentation.

Table 1 - Same-unit maintenance limits

limit_id	component	measurement_type	unit	limit_type	limit_value	record_note
LIM-001	gearbox	gearbox oil temperature	degC	maximum	82	review limit for sump temperature
LIM-002	gearbox DE bearing	vibration velocity horizontal	mm/s	maximum	6.0	route vibration review level
LIM-003	motor DE bearing	vibration velocity horizontal	mm/s	maximum	5.6	motor route vibration review level
LIM-004	motor	motor current	A	maximum	98	confirmed motor full-load current
LIM-005	mechanical seal	seal leakage rate	ml/min	maximum	4.0	timed leakage review level
LIM-006	seal flush	seal flush flow	L/min	minimum	2.5	minimum panel rotameter flow
LIM-007	seal flush	seal flush pressure	bar	acceptable range	1.5 to 2.5	panel pressure target range
LIM-008	gearbox oil	viscosity at 40C	cSt	acceptable range	300 to 340	ISO VG 320 service oil range
LIM-009	gearbox oil	water content	ppm	maximum	500	water content review level
LIM-010	gearbox oil	ISO cleanliness code	code	maximum	18/16/13	particle cleanliness review code
LIM-011	coupling	coupling horizontal offset	mm	maximum	0.08	laser alignment review level
LIM-012	coupling	coupling vertical offset	mm	maximum	0.08	laser alignment review level
LIM-013	coupling	coupling angularity	deg	maximum	0.10	laser angularity review level
LIM-014	agitator shaft	agitator shaft runout at coupling	mm	maximum	0.05	dial indicator review level
LIM-015	motor foot front left	motor foot front left soft-foot	mm	maximum	0.05	soft-foot review level
LIM-016	gearbox DE bearing	bearing temperature	degC	maximum	80	bearing contact-probe review level
LIM-017	coupling guard	largest mesh opening	mm	maximum	12	guard mesh review maximum
LIM-018	motor base	M20 motor base bolt torque	Nm	acceptable range	176 to 192	torque range
LIM-019	gearbox foot	M24 gearbox foot bolt torque	Nm	acceptable range	250 to 280	torque range
LIM-020	coupling	M10 coupling element bolt torque	Nm	acceptable range	64 to 72	torque range
LIM-021	coupling guard	M8 coupling guard screw torque	Nm	acceptable range	12 to 16	torque range
LIM-022	seal gland	M12 seal gland bolt torque	Nm	acceptable range	72 to 84	torque range

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Record-handling rules for reconciliation outputs

rule_id	rule_text	application
R-001	Blank numeric measurement values and blank required reference fields are MISSING.	Do not infer values from nearby rows or schedule values.
R-002	Descriptive notes without a numeric value and unit are TEXT-ONLY.	Examples include cloudy sight glass, dust loaded breather, powder deposits, or notes about smells.
R-003	A DUPLICATE row repeats the same source file, agitator tag, component, item description, unit, and source value.	The duplicate comparison_source_file must equal the source_file.
R-004	Gearbox oil grade in text units may be compared only with another oil-grade text record.	Do not compare oil grade against oil fill quantity.
R-005	Impeller speed tachometer readings in rpm must not be matched to calculated speed unless the item description explicitly states the same measurement method.	Calculated ratio speed and tachometer speed are distinct records.
R-006	Coupling angularity in degrees must be compared only with angularity limits in degrees.	Do not compare angularity with offset limits in millimetres.
R-007	Shaft runout in millimetres must be compared only with shaft runout limits in millimetres.	Do not compare runout with coupling offset or soft-foot values.
R-008	Seal flush flow in L/min must not be compared with seal flush pressure in bar.	Treat the flow and pressure rows as separate measurement types.
R-009	Text part-number records may be MATCH or DIFFERENT only against explicit part-number records.	A blank part-number field remains MISSING.
R-010	Use exact source filenames in source_file and comparison_source_file fields.	Abbreviated file IDs are not sufficient for the output schema.

Reference scope: This document contains internal maintenance-limit values for the AGT-44 maintenance record set.